

GrainFlow X50 Master User Guide (Enhanced Edition)

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Legal Disclaimer

Legal disclaimer for all GrainFlow X50 documentation.

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Critical Safety Warnings

DANGER: High Voltage & Electrical Shock Hazard



DANGER:

DANGER: Electrical maintenance must be performed exclusively by certified electricians. Always apply Lockout/Tagout (LOTO) procedures on the main 400V 3-phase circuit isolator before opening control enclosures.

DANGER: Rotating Drive Belts & Reciprocating Sieve Boat



DANGER:

DANGER: Entanglement hazard! Never operate the GrainFlow X50 with belt guards or inspection hatches removed. Keep hands and loose clothing clear of moving parts.

High Severity Hazards

WARNING: Airborne Grain Dust Hazard



Warning:

WARNING: Inhalation of airborne grain dust can cause respiratory illness. Always wear an approved FFP2 dust mask during cleaning and filter maintenance.

WARNING: High Acoustic Noise Emission



Warning:

WARNING: Operating sound levels reach 82 dB(A). Hearing protection must be worn at all times in the vicinity of the operating machine.

Medium Severity Operational Cautions

CAUTION: Sieve Frame Bolt Torque



CAUTION:

CAUTION: Do not over-tighten sieve frame clamping bolts beyond 45 Nm. Over-tightening can warp aluminum screen frames and void screen warranty.

CAUTION: Air Ducting Seal Integrity



CAUTION:

CAUTION: Ensure all access panel foam gaskets are free of grit before clamping. Damaged seals reduce aspiration efficiency by up to 25%.

Low Severity Technical Notes

NOTICE: Approved Synthetic Lubricants

Note:

NOTE: For maximum bearing lifespan, always use Bühler certified food-grade synthetic grease (Buhler-Lube-G2).

NOTICE: Automatic Power Saving Mode

Note:

NOTE: When idle for more than 15 minutes, the aspiration fan automatically throttles down to 30% speed to conserve electrical energy.

Product Overview

The GrainFlow X50 is a high-capacity grain cleaning system designed for industrial agricultural operations.

Introduction

The GrainFlow X50 is a state-of-the-art machine that efficiently separates impurities from grains using precision airflow and advanced screening technology. It is part of the GrainFlow Series.

Key Features

- **Throughput:** Up to 80 tonnes per hour
- **Efficiency:** 99.5% impurity removal rate
- **Energy Consumption:** 15% more efficient than previous models
- **IoT Enabled:** Remote monitoring via Bühler Connect
- **Safety:** Advanced safety interlocks and emergency stop system

Applications

The GrainFlow X50 is suitable for:

- Wheat
- Barley

- Corn/Maize
- Rice
- Soybeans
- Oats

Standard Model

The standard model offers throughput up to 50 tonnes per hour with basic IoT monitoring.

Advanced Model

The advanced model offers throughput up to 80 tonnes per hour with enhanced IoT capabilities including predictive maintenance alerts.

Operating Principles

Theory of operation for the GrainFlow X50, explaining aerodynamic aspiration, dual-deck screening, and particle separation mechanics.

Two-Stage Separation Overview

The GrainFlow X50 utilizes a high-efficiency two-stage cleaning process combining high-velocity aerodynamic aspiration with double-deck eccentric screening. Raw grain entering the machine undergoes pre-aspiration before passing over precision-perforated sieve beds, followed by final post-aspiration.

1. Pneumatic Air Separation (Aspiration)

Air separation relies on terminal velocity differences between clean grain kernels and lightweight impurities (straw, chaff, dust, and empty hulls).

- **Pre-Aspiration:** As incoming grain cascades over the distribution feed roller, a controlled air curtain extracts light dust prior to mechanical screening.
- **Post-Aspiration:** After screening, cleaned grain passes through a vertical air channel with adjustable air velocity matching the specific grain density.

2. Mechanical Sieve Deck Screening

The sieve deck operates via counterbalanced eccentric drives producing a reciprocating throw angle of 12°. Grain flows across two distinct sieve layers:

- **Scalping Sieve (Top Deck):** Retains oversized foreign particles (stones, corn cobs, large sticks) while allowing clean grain and fine impurities to drop through.
- **Sand/Fines Sieve (Bottom Deck):** Retains prime grain while allowing undersized broken kernels, weed seeds, and sand to fall through the perforations.

Premium Multi-Velocity Air Balance

On GrainFlow X50 Premium models operating at 100 TPH, automated static pressure sensors dynamically regulate the recirculation fan dampers, maintaining optimal separation efficiency across fluctuating moisture levels.

Machine Components Overview

Structural breakdown and key functional assemblies of the GrainFlow X50 grain cleaning machine.

Primary Sub-Assemblies

The GrainFlow X50 consists of five primary mechanical and electrical sub-assemblies engineered for maximum reliability under continuous duty.

1. Feed Hopper & Variable Speed Feed Roller

Accepts raw bulk grain up to 1200 L and ensures an even, single-kernel thick curtain across the entire width of the aspirator channel.

2. Reciprocating Sieve Boat

Constructed from lightweight aircraft-grade aluminum. Holds interchangeable sieve frames equipped with rubber ball cleaners to prevent screen blinding during operation.

3. Air Recirculation Separator & Blower Fan

Driven by a 45 kW motor, generating up to 14,000 m³/h airflow. Recirculates 85% of process air to minimize external dust emissions.

4. Heavy-Duty Cyclone Collector

Separates fine chaff and airborne dust from exhaust air before discharging collected waste into waste bins or central dust extraction systems.

5. Touchscreen Control Panel (Bühler YieldMaster UI)

Provides centralized control over feed rate, air damper position, sieve frequency, and real-time sensor diagnostic readouts.

Safety Features

Overview of safety systems on the GrainFlow X50.

Safety Systems

System	Description	Location
Emergency Stop	Immediately stops all moving parts	Control panel + remote points
Safety Guards	Protect operators from moving parts	All access points
Temperature Sensors	Monitor bearing and motor temperature	Bearings and motor housing
Vibration Detection	Detects imbalance and alerts operator	Main frame
Interlock System	Prevents operation when guards are open	All access doors

Emergency Stop System

Learn how the emergency stop system protects operators and equipment.

System Overview

The GrainFlow X50 is equipped with a comprehensive emergency stop system that can be activated from multiple locations.

- Main control panel emergency stop button
- Remote emergency stops at conveyor access points
- Wireless emergency stop fob (optional)

Standard Model

The standard model includes one emergency stop button on the control panel.

Advanced Model

The advanced model includes three additional remote emergency stops for improved safety coverage across the entire machine footprint.

First-Time Setup & Commissioning

Step-by-step procedure for unpacking, leveling, electrical connection, and initial calibration of the GrainFlow X50.

Prerequisites

Ensure the installation site satisfies the following requirements:

- Level concrete foundation rated for minimum 4500 kg static load.
- Power outlet matching 400V / 50 Hz / 3-phase.
- Required PPE worn: Safety glasses, hearing protection, safety footwear, dust mask (FFP2).

1. Unpack & inspect machine for transit damage.

Remove anti-corrosion transit brackets securing the sieve boat drive eccentric.

2. Level and mount the machine frame.

Place spirit level on top frame rail of sieve boat. Adjust corner leveling feet until horizontal deviation is less than 0.5 mm/m. Anchor base plates with M16 expansion bolts.

3. Connect electrical power supply.

Connect main power cables to terminal block L1, L2, L3, N, PE in accordance with local regulations. Verify rotation direction of blower motor matches directional arrow.

4. Perform sensor zero-point calibration.

From the PLC touchscreen, navigate to Service > Calibration and perform zero-point tare calibration on aspirator air pressure sensors and feed gate actuators.

5. Execute initial dry test run.

Perform a 15-minute un-loaded test run to check for abnormal vibration, bearing heat rise, or air seal leakage.

Startup Procedure

Follow these steps to safely start the GrainFlow X50.

Pre-Start Checklist

Complete the following checks before starting the machine:

Check	Action Required
Visual Inspection	Inspect all guards, covers, and safety devices for damage or missing parts
Power Supply	Verify main power supply is connected and within voltage specification (400V ± 10%)
Cleaning	Ensure all interior chambers are free from debris from the previous shift
Bearing Lubrication	Check bearing lubrication status. Top up if necessary
Emergency Stop	Verify emergency stop button is released (twist to reset)

1. Locate the main power switch on the control panel.
Refer to the control panel diagram in the appendix.
2. Turn the switch to the **ON** position.
The HMI (Human Machine Interface) will display the startup screen.
3. Wait 60 seconds for the system self-diagnostic to complete.
The system checks sensor responses, motor RPM, and fault codes.
4. Enter the desired throughput rate on the HMI touchscreen.
Default rate is 50 tonnes per hour. Range is 20-80 TPH.
5. Press the **Start** button on the HMI screen to begin operation.
The system starts main motor, conveyor belt, and cleaning process.

Adjusting Airflow Settings

Optimize grain cleaning efficiency by adjusting airflow.

Understanding Airflow

The GrainFlow X50 uses a precision airflow system to separate lighter impurities from the grain.

Low Airflow (10-20%)	Used for delicate grains to prevent damage (e.g., rice, millet)
Medium Airflow (40-60%)	Standard setting for wheat, barley, and oats
High Airflow (80-100%)	Used for heavy grains or high moisture content (e.g., maize, soybeans)

1. Access the **Settings** menu on the HMI.
2. Select **Airflow Configuration**.
3. Use the slider or numeric input to set desired airflow percentage.
Current airflow is displayed in real-time.
4. Press **Apply** to save changes.
The new airflow setting will take effect immediately.

Wheat-Specific Settings

For wheat, we recommend an airflow setting between 50-65% to achieve optimum cleaning efficiency.

Maize-Specific Settings

Maize requires higher airflow. Start at 75% and adjust based on moisture content.

Rice-Specific Settings

Rice is delicate and requires lower airflow. Recommended setting is 20-30%.

Clearing Machine Blockages & Jams

Safe procedure for isolating power, opening access hatches, and removing material blockages from the sieve boat or aspirator duct.

CRITICAL SAFETY WARNING

DANGER: NEVER insert hands or tools into the machine while power is connected. Lockout/Tagout (LOTO) is mandatory before opening access doors!

1. Press the red E-STOP button on the control panel or remote station.
2. Turn main isolator switch to OFF position and secure with personal padlock and lockout tag.
3. Wait minimum 3 minutes for sieve boat eccentric and blower fan to come to complete standstill.
4. Unclamp quick-release latches on side inspection doors.
5. Using wooden scraper or brass scraper (non-sparking), remove accumulated straw, foreign objects, or damp grain crust.
6. Inspect vertical aspiration channel and clean air distribution vanes.
7. Close all inspection doors, verify latches are locked, remove LOTO padlock, reset E-STOP, and initiate standard startup.

Shutdown Procedure

Follow these steps to safely shut down the GrainFlow X50.

1. Press the **Stop** button on the HMI.

The machine will begin its cooldown cycle (approx 30 seconds).

2. Wait for the machine to completely stop.

Do not open any guards until the machine has completely stopped.

3. Turn the main power switch to the **OFF** position.
4. Lock out/tag out the power supply if maintenance is required.

Always use lock-out/tag-out procedures before performing any maintenance or cleaning.

Post-Shutdown Checklist

- Record production data from the HMI
- Note any anomalies observed during operation
- Clean the machine if required for the next shift

Master Maintenance Schedule

Comprehensive preventative maintenance intervals for the GrainFlow X50, categorised by daily, weekly, monthly, and annual service routines.

Daily Maintenance Tasks (Every 8-10 Operating Hours)

- Inspect feed hopper for foreign objects.
- Clean aspirator glass windows and light sensors.
- Check rubber ball cleaning tray underneath sieve screens.
- **Dusty Zone Extra:** Blow out motor cooling fan shrouds with compressed air every 4 hours.

Weekly Maintenance Tasks (Every 50 Operating Hours)

- Inspect sieve deck tension bolts and torque to specified 45 Nm.
- Check drive belt tension and wear on blower fan pulley.
- Drain moisture separator on pneumatic control circuit.
- **Cold Zone Extra:** Inspect condensate drains daily to prevent frost buildup in pneumatic lines.

Monthly Maintenance Tasks (Every 200 Operating Hours)

- Grease main eccentric drive shaft bearings with Bühler Food-Grade Synthetic Grease (Buhler-Lube-G2).
- Inspect rubber flexible isolation mounts for micro-tears or degradation.
- Check electrical terminal connections inside main control cabinet.

Annual / Major Service (Every 5000 Operating Hours)

- Replace all sieve boat rubber suspension mounts.
- Calibrate motor current sensors and vibration monitors.
- Conduct full laser alignment of drive motor and eccentric shaft.

Daily Cleaning Procedure

Proper daily cleaning ensures optimal performance and longevity of the GrainFlow X50.

Safety First

Always disconnect the machine from the power supply before cleaning. Use lock-out/tag-out procedures.

1. Shut down the machine and disconnect power.
2. Open the access doors and safety guards.
3. Remove and clean the pre-filter screens.

Use compressed air to remove debris, or wash with water and mild detergent if heavily soiled.

4. Clean the interior chambers with a soft brush and vacuum.
Pay special attention to corners where dust can accumulate.
5. Wipe down the exterior with a clean, dry cloth.
6. Close all access doors and guards.
7. Reconnect power and perform a system check.

Cleaning Schedule

Component	Frequency
Pre-filter screens	Every 8 hours (or shift change)
Interior chambers	Daily
Exterior	Daily
Main filter system	Weekly

Filter Replacement

Replacing filters at the correct interval maintains airflow efficiency.

When to Replace Filters

- Every 500 operating hours
 - When airflow drops below 80% of optimal
 - Visible damage to filter material
1. Shut down the machine and disconnect power.
Never change filters while the machine is running or connected to power.
 2. Open the main filter access panel.
 3. Remove the old filter element.
Dispose of the old filter according to local environmental regulations.
 4. Clean the filter housing using a vacuum.
 5. Install the new filter element, ensuring proper alignment.
 6. Close the access panel securely.
 7. Restart the machine and verify normal airflow.

Common Issues and Solutions

Quick reference for common operational issues.

Excessive Noise from the Machine

Potential Causes:

- Loose components or guards
- Worn bearings
- Foreign object inside the chamber
- Belt tension incorrect

Solutions:

- Stop the machine immediately.
- Conduct a visual inspection of all external and accessible internal components.
- Check bearing temperature. Replace if bearings are overheating.
- If noise persists, contact Bühler Support.

Excessive Vibration During Operation**Potential Causes:**

- Machine not level
- Imbalanced rotor
- Worn vibration dampers
- Uneven feeding

Solutions:

- Check machine alignment and leveling using a precision level.
- Adjust leveling feet as needed.
- Inspect vibration dampers for wear. Replace if cracked or flattened.
- Ensure even grain flow across the entire width of the machine.

Low Throughput**Potential Causes:**

- Blocked screens or filters
- Incorrect airflow settings
- Grain moisture too high
- Motor speed incorrect

Solutions:

- Clean screens and filters
- Adjust airflow settings
- Dry grain before processing
- Verify motor RPM settings

Frequently Asked Questions (FAQ)

Quick answers to common operational, technical, and maintenance questions regarding the GrainFlow X50.

Q1: Why won't my GrainFlow X50 start?

A: Check the following checklist before calling support:

- Main electrical isolator is in the ON position and 400V power is present.
- Emergency stop button is disengaged (twist clockwise to reset).
- All safety inspection doors and belt guards are latched shut (interlocks active).
- No active critical alarm present on HMI.

Q2: How often should I clean the machine?

A: Perform daily cleaning of dust collection chambers and sieve ball trays. Weekly maintenance includes filter inspection and belt tension verification.

Q3: My throughput is lower than expected. What should I check?

A: Reduced throughput is typically caused by improper feed gate adjustment, blinded sieve perforations, or insufficient air velocity. Inspect the feed roll gap and clean the sieve screens using rubber ball agitators.

Q4: What is the maximum recommended grain moisture content?

A: The optimal moisture content for peak cleaning efficiency is below 15%. Processing grain above 18% moisture may lead to screen blinding and screen deck clumping.

Q5: How do I change the sieve screens for a different crop type?

A: Power down and lock out the machine. Release the quick-clamping levers on the sieve boat, slide out the current screen frames, insert the new crop-specific screens, and re-torque clamping bolts to 45 Nm.

Q6: How do I adjust air balance when switching from wheat to light barley?

A: Access the manual damper handwheel or HMI air control menu. Reduce aspirator fan speed or close air dampers slightly to prevent good barley kernels from being lifted into the light chaff discharge.

Q7: How do I connect the GrainFlow X50 to Bühler Connect Cloud?

A: Connect an Ethernet cable from the PLC cabinet to your plant network. Enter your Bühler IoT tenant ID in the HMI Network menu. Predictive maintenance data will start syncing automatically.

Q8: Is the machine rated for ATEX explosive dust environments?

A: Models equipped with the ATEX Option Package are certified for ATEX Zone 22 (Dust) internal and external installation in accordance with EN 1127-1.

Q9: What is covered under the standard warranty?

A: The machine is covered by a 24 months factory warranty covering structural, mechanical, and electrical component defects. Consumables (screens, belts, rubber balls) are excluded.

Q10: How can I order replacement parts or request emergency field service?

A: Contact Bühler Global Support at +41 71 955 11 11 or email support.grainflow@buhlergroup.com. Please have your machine serial number ready.

Technical Specifications

Complete technical specifications for the GrainFlow X50.

Mechanical Specifications

Parameter	Value
Dimensions (L x W x H)	2.5m x 1.8m x 3.2m
Weight	4,500 kg
Hopper Capacity	1,200 L
Power Supply	400V / 50 Hz / 3-phase
Motor Power	45 kW

Throughput Specifications

Grain Type	Standard Model	Advanced Model
Wheat	50 TPH	80 TPH

Grain Type	Standard Model	Advanced Model
Barley	45 TPH	70 TPH
Maize	40 TPH	65 TPH
Rice	35 TPH	55 TPH
Soybean	45 TPH	72 TPH
Oats	40 TPH	65 TPH

Environmental Specifications

- Operating Temperature: 5°C to 45°C
- Storage Temperature: -20°C to 60°C
- Humidity: 10% to 90% (non-condensing)
- IP Rating: IP54